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HEATCAST NCR

Can machine learning predict tomorrow's heat level in Metro Manila better than just copying today's?

CCMACLRL Introduction to Machine Learning · COM241ML · Dr. Armida P. Salazar

OUR QUESTION

One city, one day ahead

Today, Manila's heat index reached 44 °C, which is Danger.
Tomorrow can go one of three ways:



Hotter

Extreme Danger
52 °C and up



Same level

Danger again
42 to 51 °C



Cooler

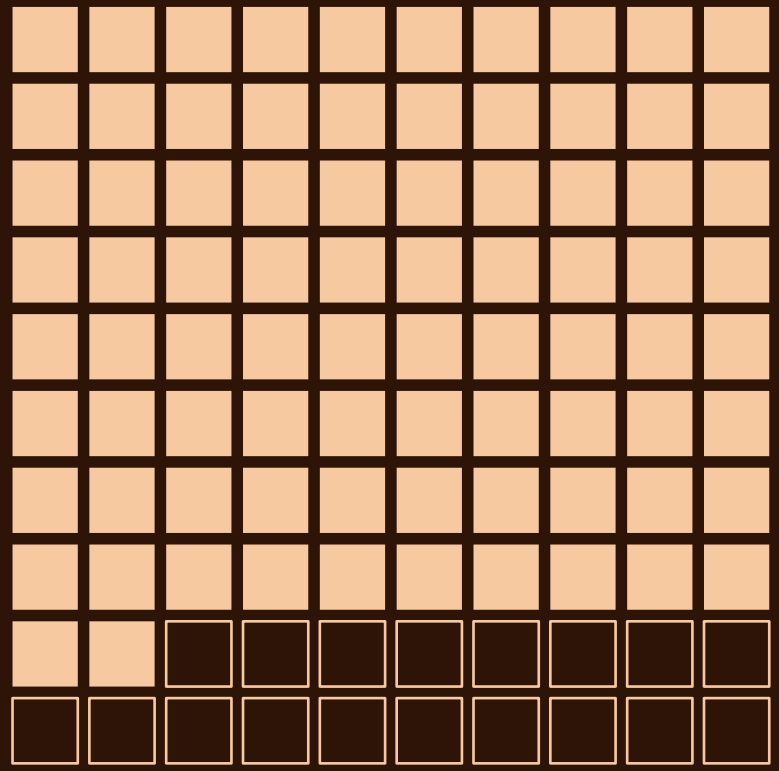
Extreme Caution or lower
below 42 °C

HeatCast NCR picks one of these for each of the 16 cities, every day.

82%

of the time in 2025, tomorrow's heat level was the same as today's.

So simply copying today's level is already a strong forecast. We call it the copy-today rule, and it is the rule our models had to beat.



Filled: same level as the day before
Empty: the level changed

THE PROBLEM AND THE DATA

Who needs a one-day heat warning, and what we used

Who needs it

Schools, employers of outdoor workers and city offices. A day ahead, they decide to move activities, prepare water and cooling, or warn residents.

Our task

Predict tomorrow's PAGASA heat level for each city, using only today's weather in that city.

The PAGASA heat index levels (the heat index is how hot it feels)

Not Hazardous	Caution	Extreme Caution	Danger	Extreme Danger
below 27 °C	27–32 °C	33–41 °C	42–51 °C	52 °C and up

16

cities of Metro Manila (NCR)

29,168

rows: one per city per day, 2021 to 2025

5

weather readings about today: heat index, lowest temperature, rain, air pressure, sunshine

1 day

ahead: we predict tomorrow's level in the same city

Weather: hourly estimates from Open-Meteo (ERA5)

WHAT THE DATA SHOWED

Three findings, and what each one changed

8%

of city-days were Danger, and 78.5% were Extreme Caution

So "percent correct" misleads: always guessing Extreme Caution is already 78% right.

May

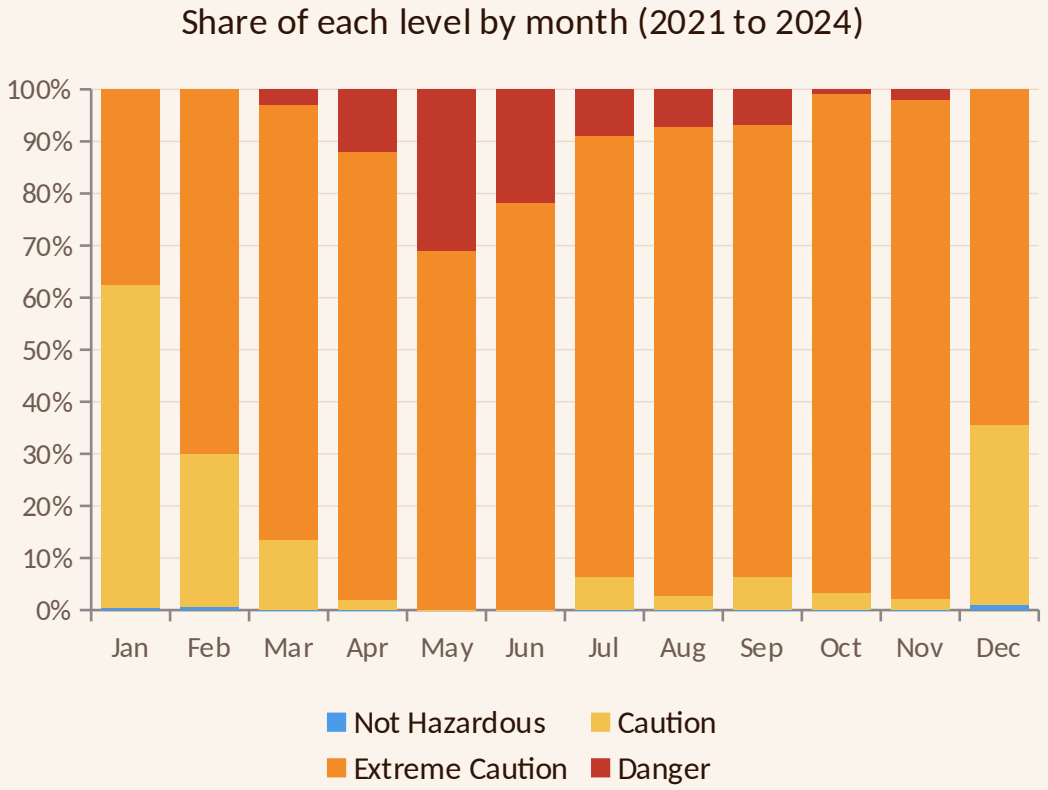
Danger peaks in April to June (nearly 1 in 3 May days) and never happens December to February

So we split the data by time, like real forecasting.

17% vs 2%

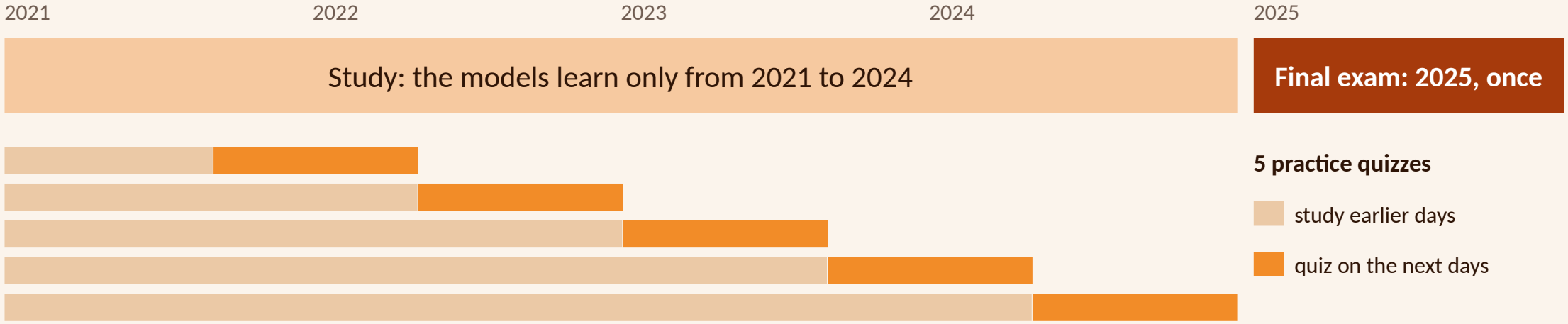
Some cities get far more Danger days: Malabon, Navotas, Valenzuela vs Muntinlupa, San Juan

So every city keeps its own rows, and the city is an input.



A FAIR TEST, LIKE AN EXAM

Every model studied the same days and took the same final exam



The score

How well a model does on Caution, Extreme Caution and Danger, each counted equally (0 to 1, higher is better).

The rule

First, catch more Danger days than the copy-today rule. Then the best score wins.

The baseline

Copy today: tomorrow's level is today's level in the same city.

In technical terms: chronological split, 5-fold time-series cross-validation, macro F1.

WHICH MISTAKE IS WORSE?

Every forecast is sometimes wrong, in one of two ways



WORSE

Missing a real Danger day

Nobody prepares, and students play outside at 44 °C.

The cost: someone may get heat exhaustion or heat stroke.



LESS COSTLY

A false alarm

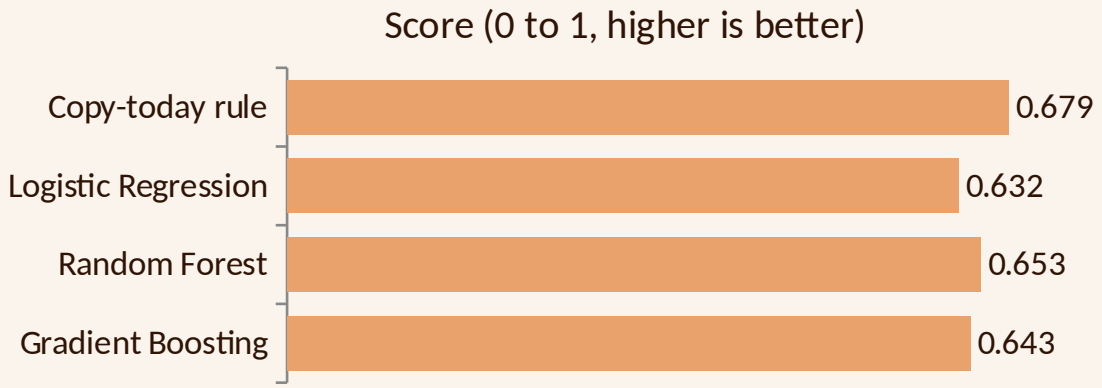
Activities are moved indoors, but the day was not that hot.

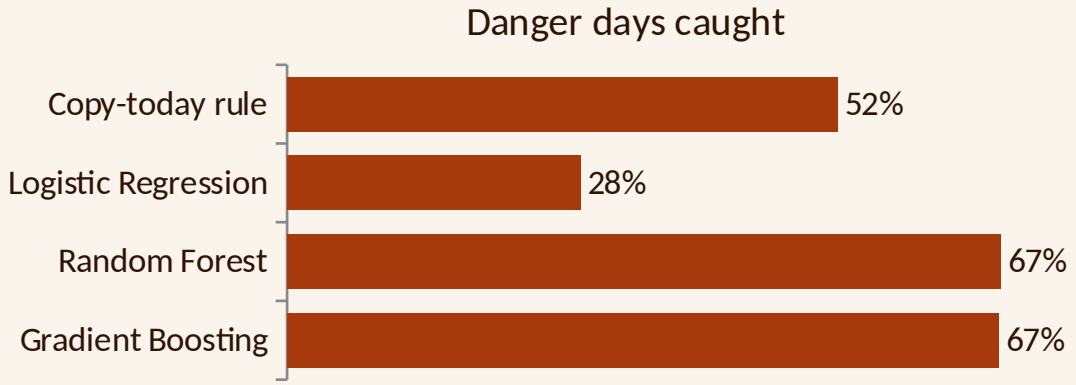
The cost: a change of plans for nothing.

So our rule: a model must first catch more Danger days than copy-today.

THREE MODELS VS. COPY-TODAY

Practice-quiz results: the average of the 5 quizzes





Logistic Regression Harry

Draws straight dividing lines between the levels.

Careful, but caught only 28% of Danger days.

Random Forest Edrian

SELECTED

500 decision trees that vote on the answer.

Caught 67% of Danger days with the best score of the three.

Gradient Boosting Marc

Trees built one after another, each fixing the last one's mistakes.

Almost tied with Random Forest.

DANGER DAYS CAUGHT

The final exam: 2025, opened once

In 2025 there were 347 Danger days across the 16 cities. Here is how many got a warning the day before.



86 more Danger days got a warning the day before, compared with copying today.

FINAL EXAM RESULT

The selected Random Forest on 2025, compared with copy-today

293

of 347 Danger days caught (84%)

Copy-today caught 207 (60%)

36%

of its Danger warnings were right

293 of 816: about 2 false alarms for each real Danger day

0.61

score, below copy-today's 0.68

It did not beat copy-today overall

Not a better all-round forecaster, but a good early warning: it catches far more Danger days than copying today, at the price of more false alarms.

LIVE DEMO

heatcast-ncr.streamlit.app

1

A hot day in Manila

We choose Manila and type a hot day: heat index 45.5 °C, lowest 29 °C, no rain, 1,008 hPa, sunshine 7,000.

2

A mistake on purpose

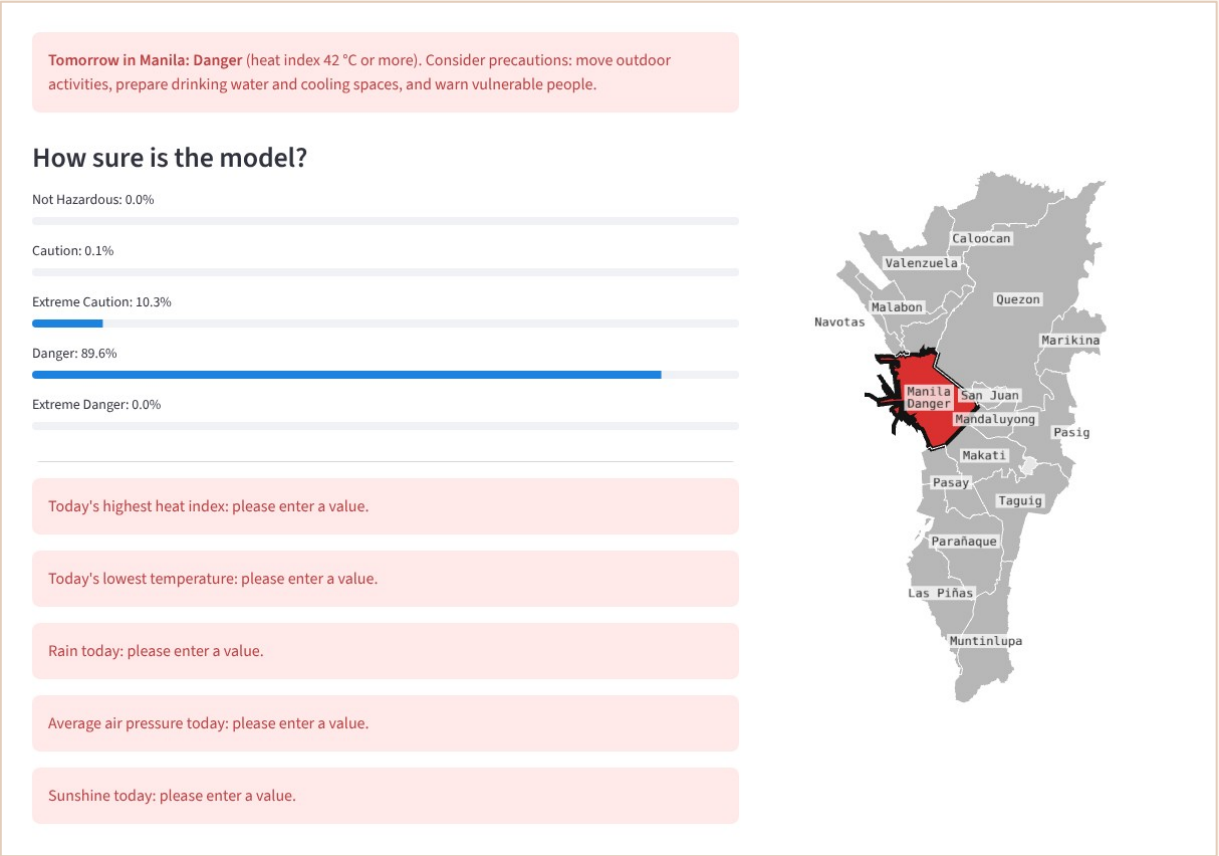
Rain of −5 mm is impossible: the app shows the accepted range and gives no prediction.

3

An empty form

Every missing value is named by the app; nothing is filled in silently.

The app uses the same saved Random Forest that took the final exam.



CONCLUSION

What we learned

01



The answer

No. None of our models beat copying today's level on overall score: tomorrow's heat is mostly today's heat.

02



What ML adds

Told that Danger days matter more, the Random Forest caught 84% of them in 2025. We deployed it as an early warning.

03



Limits and next steps

Weather estimates, not station readings; only five years. Next: add the month and recent days, a weather forecast, and more years.

THANK YOU

Questions?

Try it: heatcast-ncr.streamlit.app · Code: github.com/keyyum/heatcast-ncr-simple

BACKUP: KEY TERMS

For questions only; we do not present this slide

Word on the slides	What it means
Heat index	How hot it feels, from temperature and humidity (US National Weather Service formula).
Copy-today rule	The persistence baseline: tomorrow's level = today's level in the same city.
Score (macro F1)	F1 for each level, averaged over Caution, Extreme Caution and Danger; each level counts equally.
Danger days caught	Danger recall: of the real Danger days, the share the model called Danger.
Warnings that were right	Danger precision: of the model's Danger calls, the share that really were Danger.
Told Danger matters more	Balanced class weights: rare levels count more while the model learns.
Practice quizzes	5-fold time-series cross-validation inside 2021 to 2024; never learns from later days.
Weather estimates	ERA5 reanalysis via Open-Meteo: a grid of about 25 km, not station readings.